GED Math Practice Test 9

PART A — QUESTIONS

GED-MATH9-001
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A bus has 48 seats. If 3/4 of the seats are filled, how many seats are filled?
A) 12
B) 24
C) 36
D) 44

GED-MATH9-002
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
What is the value of $5(2x - 1)$ when $x = 4$?
A) 25
B) 30
C) 35
D) 45

GED-MATH9-003
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A shelf contains 18 paperback books and 12 hardcover books. What percent of the books are hardcover?
A) 30%
B) 40%
C) 50%
D) 60%

GED-MATH9-004
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
Which expression is equivalent to $4(3a - 2) - 7a$?
A) $5a - 8$
B) $5a + 8$

C) $12a - 9$
D) $19a - 8$

GED-MATH9-005
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A recipe needs 1/2 cup of oil for 4 servings. How many cups of oil are needed for 12 servings at the same rate?
A) 1 cup
B) 1 1/2 cups
C) 2 cups
D) 3 cups

GED-MATH9-006
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $D = 1.2t^2$ gives the distance D, in meters, an object rolls down a ramp after t seconds. If $D = 30$, what is t?
A) 5
B) 10
C) 25
D) 36

GED-MATH9-007
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A storage container holds 36 small boxes. Each small box has 24 packets, and each packet weighs 0.75 ounce. What is the total weight, in ounces?
Type your answer as an integer.

GED-MATH9-008
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $P = 2l + 2w$ gives the perimeter P of a rectangle. To solve this formula for l, subtract [D1] from both sides, then divide by [D2].
Dropdowns:
D1) 2w | 2l | P
D2) 2 | w | P

GED-MATH9-009
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A bag of rice weighs 4.8 kilograms. How many grams does the bag weigh?
A) 48 grams
B) 480 grams
C) 4,800 grams
D) 48,000 grams

GED-MATH9-010
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A value of n satisfies $7 - 2n > -9$. Which value of n does NOT satisfy the inequality?
A) -4
B) 0
C) 7
D) 9

GED-MATH9-011
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A data plan allows 15 gigabytes per month. A user has used 9.6 gigabytes. What percent of the monthly data has been used?
A) 56%
B) 60%
C) 64%
D) 72%

GED-MATH9-012
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
Which expressions are equivalent to $18 - 6x$?
A) $6(3 - x)$
B) $-6(x - 3)$
C) $3(6 - x)$
D) $6(x - 3)$
E) $-6x - 18$

GED-MATH9-013
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows how many containers of each size are in a cafeteria storage room.

Container size: 1 quart, 2 quart, 5 quart
Number of containers: 12, 9, 6

What is the total storage capacity of all the containers?
A) 27 quarts
B) 36 quarts
C) 60 quarts
D) 90 quarts

GED-MATH9-014
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $H = 0.04d + 1.5$ gives the height H, in feet, of a young tree after d days of growth. If $H = 4.7$, what is d?
Type your answer as an integer.

GED-MATH9-015
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A composite shape is made from a rectangle and a triangle. The rectangle is 14 inches long and 9 inches wide. The triangle has a base of 14 inches and a height of 6 inches. What is the total area of the composite shape?
A) 84 square inches
B) 126 square inches
C) 168 square inches
D) 210 square inches

GED-MATH9-016
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a linear relationship.

x: 3, 7, 15, 19
y: −2, 10, 34, 46

Which equation represents the relationship?
A) $y = 2x - 8$
B) $y = 3x - 11$
C) $y = 4x - 14$
D) $y = 6x - 20$

GED-MATH9-017
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A store compared two sizes of the same detergent.

Size: Regular, Family
Ounces: 48, 80
Price: $6.24, $9.20

Which statements are true?
A) The regular size costs $0.13 per ounce.
B) The family size costs $0.115 per ounce.
C) The regular size has the lower unit price.
D) The family size costs $0.13 per ounce.
E) The two sizes have the same unit price.

GED-MATH9-018
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which inequality represents “the product of 4 and the sum of a number and 3 is less than 52”?
A) $4x + 3 < 52$
B) $4(x + 3) < 52$
C) $4(x - 3) < 52$
D) $x + 4(3) < 52$

GED-MATH9-019
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A jacket is priced at $96 after a 25% markup from the store’s cost. What was the store’s cost?
A) $72.00
B) $76.80
C) $84.00
D) $120.00

GED-MATH9-020
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for x: $6(x - 4) - 2(x + 5) = 30$.
Type your answer as an integer.

GED-MATH9-021
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A circular garden has a diameter of 14 feet. Using $\pi \approx 3.14$, what is the area of the garden?
A) 43.96 square feet
B) 87.92 square feet
C) 153.86 square feet
D) 615.44 square feet

GED-MATH9-022
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A line passes through the points $(-5, 18)$ and $(1, 0)$. What is the rate of change of the line?
A) -4
B) -3
C) 3
D) 4

GED-MATH9-023
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A runner completes 9.3 miles on Saturday and 6.8 miles on Sunday. The runner’s goal is to complete 20 miles total. How many more miles are needed?
A) 3.1 miles
B) 3.9 miles
C) 4.1 miles
D) 6.1 miles

GED-MATH9-024
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed

Prompt:
Function A is represented by the table.

x: 1, 4, 7, 10
y: 24, 18, 12, 6

Function B is represented by $y = -3x + 28$. Which statement is true?
A) Function A decreases faster than Function B.
B) Function B decreases faster than Function A.
C) Both functions have the same rate of change.
D) Function A has a positive rate of change.

GED-MATH9-025
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A course grade is based on quizzes, labs, and a final exam. Quizzes count 20%, labs count 35%, and the final exam counts 45%. A student scores 88 on quizzes, 94 on labs, and 76 on the final exam. What is the course grade?
Type your answer as a decimal rounded to the nearest tenth.

GED-MATH9-026
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A parking garage collected money from 64 vehicles. Cars paid $8 each, and trucks paid $13 each. The garage collected $612 total. Which system can be used to find c, the number of cars, and t, the number of trucks?
A) $c + t = 612$
$8c + 13t = 64$
B) $c + t = 64$
$8c + 13t = 612$
C) $8c + 13t = 64$
$c - t = 612$
D) $13c + 8t = 64$
$c + t = 612$

GED-MATH9-027
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A square room has a perimeter of 52 feet. What is the area of the room?
A) 13 square feet
B) 52 square feet

C) 169 square feet
D) 676 square feet

GED-MATH9-028
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The equation $4x - 5y = 30$ can be rewritten as y = [D1]x + [D2].
Dropdowns:
D1) 4/5 | −4/5 | 5/4
D2) −6 | 6 | 30

GED-MATH9-029
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A drawer contains 3 black socks, 5 gray socks, and 4 white socks. One sock is selected at random, replaced, and then another sock is selected. What is the probability that both socks are black?
A) 1/16
B) 1/12
C) 1/9
D) 1/4

GED-MATH9-030
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The formula A = 1/2bh gives the area A of a triangle. Which equations correctly solve for b?
A) b = 2A/h
B) b = A/(2h)
C) b = 2Ah
D) b = A ÷ (1/2h)
E) b = h/(2A)

GED-MATH9-031
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
An object has a mass of 1,080 grams and a volume of 135 cubic centimeters. What is the density of the object?
A) 6 g/cm^3
B) 8 g/cm^3

C) 10 g/cm^3
D) 12 g/cm^3

GED-MATH9-032
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for r: $5r - 2(3r - 9) = 41$.
A) −23
B) −13
C) 13
D) 23

GED-MATH9-033
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A cone has a radius of 4 inches and a height of 15 inches. Using $\pi \approx 3.14$, what is the volume of the cone?
Use $V = 1/3\pi r^2h$.
A) 62.8 in^3
B) 188.4 in^3
C) 251.2 in^3
D) 753.6 in^3

GED-MATH9-034
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A quadratic expression is $x^2 - 9x + 20$. Which value of x makes the expression equal to 0?
A) 2
B) 4
C) 9
D) 20

GED-MATH9-035
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A small town's population is modeled by P = 12,500(0.97)^t, where t is the number of years after the first measurement. What does 0.97 mean in this model?
A) The population decreases by 3% each year.
B) The population increases by 3% each year.

C) The starting population is 97 people.
D) The population decreases by 97 people each year.

GED-MATH9-036
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
On a scale drawing, 2.5 centimeters represents 18 meters. A hallway is 9 centimeters long on the drawing. What is the actual length of the hallway?
A) 45.0 meters
B) 54.0 meters
C) 64.8 meters
D) 72.0 meters

GED-MATH9-037
Section: Math
Type: drag_drop
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Match each target with the correct tile.
Tiles:
A) $2(x + y) = 40$
B) $2x + y = 40$
C) $x + 2y = 40$
D) $2(x - y) = 40$
Targets:
T1) Twice the sum of two numbers is 40
T2) Twice the first number plus the second number is 40
T3) The first number plus twice the second number is 40

GED-MATH9-038
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A machine uses 2.4 kilowatt-hours of electricity in 3 hours. At the same rate, how many kilowatt-hours will it use in 11 hours?
A) 6.6
B) 8.8
C) 13.2
D) 26.4

GED-MATH9-039
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed

Prompt:
Which ordered pair satisfies the inequality $y \geq 3x - 2$?
A) (1, 0)
B) (2, 3)
C) (3, 8)
D) (4, 8)

GED-MATH9-040
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A value is modeled by V = 240(1.15)^t. Which statement best describes the change in V each time t increases by 1?
A) V increases by 15%.
B) V decreases by 15%.
C) V increases by 115%.
D) V decreases by 85%.

GED-MATH9-041
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The table shows weekly earnings for two workers.

Worker: Mia, Noah
Hours worked: 32, 38
Total earnings: $544, $646

Which statements are true?
A) Mia earned $17 per hour.
B) Noah earned $17 per hour.
C) Noah earned $19 per hour.
D) Mia earned more per hour than Noah.
E) Together, they earned $18 per hour.

GED-MATH9-042
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
For the function $g(x) = -x^2 + 4x + 12$, what is $g(-3)$?
A) −9
B) −3
C) 15
D) 33

GED-MATH9-043
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a relationship.

x: 0, 1, 2, 3
y: 5, 8, 13, 20

Which statement best describes the relationship?
A) It is linear because y increases each time.
B) It is linear because x increases by 1 each time.
C) It is not linear because the changes in y are not constant.
D) It is not linear because the first y-value is not 0.

GED-MATH9-044
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which expression represents 7 less than three times the sum of a number x and 4?
A) $3x + 4 - 7$
B) $3(x + 4) - 7$
C) $7 - 3(x + 4)$
D) $3(x - 4) + 7$

GED-MATH9-045
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $V = lwh$ gives the volume of a rectangular prism. If $V = 1{,}440$, $l = 15$, and $w = 8$, what is h?
Type your answer as an integer.

GED-MATH9-046
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular pool is 25 feet long and 14 feet wide. A concrete border 3 feet wide surrounds the pool. What is the area of the concrete border?
A) 126 square feet
B) 270 square feet
C) 350 square feet
D) 584 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH9-001 — Correct: C — Correct Answer: 36 — Explanation: Three fourths of 48 is $48 \times 3/4 = 36$.

GED-MATH9-002 — Correct: C — Correct Answer: 35 — Explanation: Substitute $x = 4$: $5(2(4) - 1) = 5(7) = 35$.

GED-MATH9-003 — Correct: B — Correct Answer: 40% — Explanation: There are 30 books total, and 12 are hardcover. Then $12/30 = 0.40$, or 40%.

GED-MATH9-004 — Correct: A — Correct Answer: $5a - 8$ — Explanation: Distribute and combine like terms: $4(3a - 2) - 7a = 12a - 8 - 7a = 5a - 8$.

GED-MATH9-005 — Correct: B — Correct Answer: 1 1/2 cups — Explanation: Twelve servings is 3 times 4 servings, so $3 \times 1/2 = 1\ 1/2$ cups.

GED-MATH9-006 — Correct: A — Correct Answer: 5 — Explanation: $30 = 1.2t^2$, so $25 = t^2$ and $t = 5$.

GED-MATH9-007 — Correct: 648 — Tolerance: exact — Correct Answer: 648 — Explanation: There are $36 \times 24 = 864$ packets. Each packet weighs 0.75 ounce, so the total is $864 \times 0.75 = 648$ ounces.

GED-MATH9-008 — Correct: D1=2w; D2=2 — Correct Answer: D1=2w; D2=2 — Explanation: From $P = 2l + 2w$, subtract $2w$ to get $P - 2w = 2l$, then divide by 2.

GED-MATH9-009 — Correct: C — Correct Answer: 4,800 grams — Explanation: One kilogram is 1,000 grams, so 4.8 kilograms is 4,800 grams.

GED-MATH9-010 — Correct: D — Correct Answer: 9 — Explanation: Solve $7 - 2n > -9$. Subtract 7 to get $-2n > -16$, so $n < 8$. The value 9 does not satisfy the inequality.

GED-MATH9-011 — Correct: C — Correct Answer: 64% — Explanation: $9.6 \div 15 = 0.64$, or 64%.

GED-MATH9-012 — Correct: A,B — Correct Answer: $6(3 - x) \mid -6(x - 3)$ — Explanation: Both expressions simplify to $18 - 6x$.

GED-MATH9-013 — Correct: C — Correct Answer: 60 quarts — Explanation: Total capacity is $12(1) + 9(2) + 6(5) = 12 + 18 + 30 = 60$ quarts.

GED-MATH9-014 — Correct: 80 — Tolerance: exact — Correct Answer: 80 — Explanation: $4.7 = 0.04d + 1.5$. Subtract 1.5 to get $3.2 = 0.04d$, so $d = 80$.

GED-MATH9-015 — Correct: C — Correct Answer: 168 square inches — Explanation: Rectangle area is $14 \times 9 = 126$. Triangle area is $1/2(14)(6) = 42$. Total area is $126 + 42 = 168$.

GED-MATH9-016 — Correct: B — Correct Answer: $y = 3x - 11$ — Explanation: When x increases by 4, y increases by 12, so the rate is 3. Substituting $x = 3$ and $y = -2$ gives $y = 3x - 11$.

GED-MATH9-017 — Correct: A,B — Correct Answer: The regular size costs \$0.13 per ounce. | The family size costs \$0.115 per ounce. — Explanation: Regular unit price: $6.24 \div 48 = \$0.13$ per ounce. Family unit price: $9.20 \div 80 = \$0.115$ per ounce.

GED-MATH9-018 — Correct: B — Correct Answer: $4(x + 3) < 52$ — Explanation: “The sum of a number and 3” is $x + 3$. The product of 4 and that sum is $4(x + 3)$.

GED-MATH9-019 — Correct: B — Correct Answer: \$76.80 — Explanation: A 25% markup means the selling price is 125% of cost. So $96 \div 1.25 = 76.80$.

GED-MATH9-020 — Correct: 16 — Tolerance: exact — Correct Answer: 16 — Explanation: $6(x - 4) - 2(x + 5) = 6x - 24 - 2x - 10 = 4x - 34$. Then $4x - 34 = 30$, so $x = 16$.

GED-MATH9-021 — Correct: C — Correct Answer: 153.86 square feet — Explanation: The radius is 7 feet. Area $= \pi r^2 = 3.14(7^2) = 153.86$ square feet.

GED-MATH9-022 — Correct: B — Correct Answer: −3 — Explanation: Rate of change $= (0 - 18)/(1 - (-5)) = -18/6 = -3$.

GED-MATH9-023 — Correct: B — Correct Answer: 3.9 miles — Explanation: The runner completed $9.3 + 6.8 = 16.1$ miles. The remaining distance is $20 - 16.1 = 3.9$ miles.

GED-MATH9-024 — Correct: B — Correct Answer: Function B decreases faster than Function A. — Explanation: Function A has rate $(18 - 24)/(4 - 1) = -2$. Function B has rate -3, so Function B decreases faster.

GED-MATH9-025 — Correct: 84.7 — Tolerance: exact — Correct Answer: 84.7 — Explanation: Course grade $= 0.20(88) + 0.35(94) + 0.45(76) = 17.6 + 32.9 + 34.2 = 84.7$.

GED-MATH9-026 — Correct: B — Correct Answer: $c + t = 64$
$8c + 13t = 612$ — Explanation: The first equation represents the number of vehicles, and the second represents the total money collected.

GED-MATH9-027 — Correct: C — Correct Answer: 169 square feet — Explanation: A square with perimeter 52 has side length $52 \div 4 = 13$ feet. Its area is $13^2 = 169$ square feet.

GED-MATH9-028 — Correct: D1=4/5; D2=−6 — Correct Answer: D1=4/5; D2=−6 — Explanation: $4x - 5y = 30$ gives $-5y = -4x + 30$, so $y = 4/5x - 6$.

GED-MATH9-029 — Correct: A — Correct Answer: 1/16 — Explanation: There are 12 socks total, and 3 are black. The probability of black is $3/12 = 1/4$ each time. With replacement, both black is $1/4 \times 1/4 = 1/16$.

GED-MATH9-030 — Correct: A,D — Correct Answer: $b = 2A/h$ | $b = A \div (1/2h)$ — Explanation: From $A = 1/2bh$, divide both sides by $1/2h$. This is equivalent to multiplying by 2 and dividing by h.

GED-MATH9-031 — Correct: B — Correct Answer: 8 g/cm^3 — Explanation: Density = mass ÷ volume $= 1{,}080 \div 135 = 8$ g/cm^3.

GED-MATH9-032 — Correct: A — Correct Answer: −23 — Explanation: $5r - 2(3r - 9) = 5r - 6r + 18 = -r + 18$. Then $-r + 18 = 41$, so $-r = 23$ and $r = -23$.

GED-MATH9-033 — Correct: C — Correct Answer: 251.2 in^3 — Explanation: $V = 1/3\pi r^2h = 1/3(3.14)(4^2)(15) = 251.2$ cubic inches.

GED-MATH9-034 — Correct: B — Correct Answer: 4 — Explanation: $x^2 - 9x + 20$ factors as $(x - 4)(x - 5)$, so $x = 4$ makes the expression equal 0.

GED-MATH9-035 — Correct: A — Correct Answer: The population decreases by 3% each year. — Explanation: Multiplying by 0.97 means 97% remains each year, so the population decreases by 3% each year.

GED-MATH9-036 — Correct: C — Correct Answer: 64.8 meters — Explanation: The scale is $18 \div 2.5 = 7.2$ meters per centimeter. Then $9 \times 7.2 = 64.8$ meters.

GED-MATH9-037 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=$2(x + y) = 40$; T2=$2x + y = 40$; T3=$x + 2y = 40$ — Explanation: Each equation matches the target's wording and grouping.

GED-MATH9-038 — Correct: B — Correct Answer: 8.8 — Explanation: The rate is $2.4 \div 3 = 0.8$ kilowatt-hour per hour. In 11 hours, the machine uses $0.8 \times 11 = 8.8$ kilowatt-hours.

GED-MATH9-039 — Correct: C — Correct Answer: (3, 8) — Explanation: Substitute (3, 8): $3(3) - 2 = 7$, and $8 \geq 7$ is true.

GED-MATH9-040 — Correct: A — Correct Answer: V increases by 15%. — Explanation: Multiplying by 1.15 means the value increases by 15% each time t increases by 1.

GED-MATH9-041 — Correct: A,B — Correct Answer: Mia earned $17 per hour. | Noah earned $17 per hour. — Explanation: Mia: $544 \div 32 = 17$. Noah: $646 \div 38 = 17$.

GED-MATH9-042 — Correct: A — Correct Answer: −9 — Explanation: $g(-3) = -(-3)^2 + 4(-3) + 12 = -9 - 12 + 12 = -9$.

GED-MATH9-043 — Correct: C — Correct Answer: It is not linear because the changes in y are not constant. — Explanation: The y-values increase by 3, then 5, then 7, so the relationship is not linear.

GED-MATH9-044 — Correct: B — Correct Answer: $3(x + 4) - 7$ — Explanation: The sum of x and 4 is $x + 4$. Three times that sum is $3(x + 4)$, and 7 less than that is $3(x + 4) - 7$.

GED-MATH9-045 — Correct: 12 — Tolerance: exact — Correct Answer: 12 — Explanation: $1{,}440 = 15(8)h$, so $1{,}440 = 120h$ and $h = 12$.

GED-MATH9-046 — Correct: B — Correct Answer: 270 square feet — Explanation: The outside dimensions are 31 feet by 20 feet, so the outside area is 620 square feet. The pool area is $25 \times 14 = 350$ square feet. The border area is $620 - 350 = 270$ square feet.